Long-Term Stockpile Reconstruction and Block Modelling

Term Project

Course	MIN E 413
Instructor	Yashar Pourrahimian
Team Size	5 students
Project Weight	20%
Due Date	Thursday, November 26 by 23:59
Submission	.Zip file on Canvas

Project at a Glance

Engineering challenge: Reconstruct the historical movement of truck-dumped material through a growing copper-gold stockpile and create a defensible three-dimensional block model of tonnage, grade, contained metal, and uncertainty.

Area	What you are expected to produce
Data analysis	QA/QC, temporal-spatial reconciliation, visualization, and documented preprocessing.
Material-movement model	A justified algorithm for dozer redistribution and gravity/down-slope movement.
Stockpile block model	3D estimates of tonnage, Cu grade, Au grade, contained metal, and uncertainty.
Validation	Geometry, mass balance, metal balance, temporal checks, and quantitative performance measures.
Automation	A reproducible in-house procedure/software platform that can process future data.
Professional practice	Transparent assumptions, limitations, AI/software disclosure, accountability, and equity considerations.
Teamwork	Team charter, contribution record, peer/self-assessment, individual reflection, and technical defence.

1. Project Context

You are a member of the Long-Term Mine Planning Team at a large copper-gold surface mine. One of the mine's open pits, Pit I, is approaching the end of its operating life and is expected to be exhausted by the end of 2026. During Pit I operations, a significant quantity of mineralized material has been placed in a long-term stockpile for processing at a later date.

Once mining from Pit I is complete, this stockpile will become an important feed source for the processing plant. A reliable understanding of the stockpile is therefore required to support future reclaim sequencing, ore blending, processing schedules, metal-recovery forecasting, and economic planning.

Unlike an in-situ geological deposit, the stockpile cannot be reconstructed simply from original truck-dumping locations. Haul trucks place discrete loads at recorded coordinates on or near the active stockpile surface. Dozers subsequently push and redistribute the material; gravity may cause additional downslope movement before the material becomes incorporated into the evolving

stockpile. The final location of a load may therefore differ substantially from its reported dump coordinate.

Mine Manager request: Design a defensible platform that reconstructs historical stockpile growth, estimates where dumped material ultimately resides, and generates an updatable stockpile block model.

2. Primary Engineering Problem

How can historical truck-dumping records and sequential stockpile surfaces be used to reconstruct material movement and final placement, and thereby generate a three-dimensional stockpile block model containing estimates of tonnage, copper grade, gold grade, and uncertainty?

The central technical challenge is to infer an unobserved process: how material initially dumped by haul trucks is redistributed by dozer activity and gravity between consecutive stockpile surveys. There is no single prescribed algorithm. Your team must formulate, justify, test, validate, and critically evaluate an engineering approach using the supplied data.

3. Available Information

3.1 Sequential Topographical Data

Detailed stockpile surface surveys are available at regular intervals, generally daily. Each surveyed surface represents the stockpile's physical configuration after dumping and dozer activities during the preceding operating period.

3.2 Fleet Management System (FMS) Data

- Truck identification, where available
- Dump coordinates (x, y, and/or elevation as available)
- Timestamp
- Dumped tonnage
- Source location or material identifier, where available
- Copper grade
- Gold grade
- Other relevant material attributes included in the supplied dataset

3.3 Current Stockpile Surface

A surveyed representation of the current stockpile geometry is available and should be used as a principal reconciliation target for the final reconstruction.

3.4 Supporting Information

You may supply other information relevant to stockpile construction, material properties, operating practices, or equipment activity. Your team is responsible for determining whether and how to incorporate such information.

4. Project Objectives

1. Analyze and quality-check the historical data before modelling.

2. Reconstruct the growth of the stockpile through time from sequential surveys and FMS records.

3. Develop an algorithm that estimates material movement from the truck dump location to the volume represented by the subsequent stockpile surface.

4. Preserve appropriate mass and metal balances during material redistribution.

5. Estimate the three-dimensional distribution of material within the current stockpile.

6. Populate a stockpile block model with tonnage, copper grade, gold grade, contained metal, and uncertainty information.

7. Validate the methodology quantitatively and identify its principal limitations.

8. Evaluate the sensitivity of results to important assumptions and uncertain parameters.

9. Develop a reproducible in-house procedure/software platform that can update the analysis when future data become available.

10. Demonstrate professional accountability, transparent use of AI/software tools, effective teamwork, and consideration of equity and inclusive engineering practice.

5. Required Project Tasks

Task 1 - Data Investigation and Quality Assessment

Before developing the material-movement algorithm, investigate the supplied datasets and determine whether they are sufficiently complete, consistent, and aligned for modelling.

- Spatial and temporal coverage of each dataset.
- Coordinate reference systems, datums, units, and transformations.
- Missing, duplicate, or inconsistent records.
- Anomalous or physically implausible dump locations.
- Unrealistic or inconsistent tonnages.
- Missing or anomalous copper and gold grades.
- Consistency between truck-dumping records and surveyed surface changes.
- Survey resolution, apparent survey noise, and possible artefacts.
- Temporal alignment between FMS records and survey timestamps.
- Any additional data-quality issue that could materially affect the reconstruction.

Develop appropriate visualizations. Useful examples may include dump-location maps, daily tonnage and grade plots, cumulative placement curves, surface-difference maps, spatial density of dumping locations, and identification of anomalous records.

Required engineering judgment: Do not simply delete questionable data. For each significant cleaning, exclusion, correction, or imputation rule, explain why it is necessary, what action was taken, how the action may influence the resulting model, and how the original information is retained for traceability.

Task 2 - Characterization of Stockpile Growth

Use the sequential topographical surfaces to determine how the stockpile evolved through time. For consecutive surfaces S_t and S_{t+1}, investigate the spatial distribution of elevation increase, elevation decrease, apparent material accumulation, apparent removal/redistribution, and changes that may reflect survey uncertainty rather than true material movement.

Compare measured geometric changes with truck tonnage reported during the corresponding period. Where required, justify material density, compaction, swell, or volume-conversion assumptions.

FMS-reported material <-> Material represented by surveyed geometric change

Significant discrepancies must be quantified, investigated, and discussed. The analysis should distinguish physical explanations from data or modelling limitations.

Task 3 - Development of the Material-Movement Algorithm

This is the core technical task. For each relevant time interval, your algorithm must estimate how material moves from its initial truck dump location to the volume represented by the subsequent stockpile surface.

Truck Dump Location -> Dozer Redistribution -> Gravity / Downslope Movement -> Estimated Final Location

You may use any reasonable computational methodology. Potential concepts include local topographic gradients, steepest-descent rules, angle-of-repose constraints, neighbourhood/cellular movement, cellular automata, volume balancing, particle methods, optimization, probabilistic allocation, machine-learning-assisted methods, or hybrid physics/data-driven approaches. These examples are directions for exploration, not prescribed solutions.

Required question	Your methodology must explain...
A. Initial placement	How the tonnage of each truck load is represented computationally at its reported dump location.
B. Direction of movement	How the model determines the direction in which material is pushed or allowed to move.
C. Distance of movement	What controls the allowable movement distance and when movement stops.
D. Downslope behaviour	How movement on stockpile faces and gravity-driven redistribution are represented.
E. Material mixing	How the algorithm combines tonnage and grades when material from different loads, times, or sources occupies the same region.
F. Surface reconciliation	How the reconstructed volume is constrained by or compared with the observed subsequent survey surface.
G. Mass conservation	How the algorithm prevents unintended creation or loss of material.

Required question	Your methodology must explain...
H. Metal conservation	How copper and gold inventories are preserved through redistribution and mixing.
I. Sequential updating	How one period becomes the initial condition for the next period.
J. Numerical stability	How grid/block resolution, time step, thresholds, or convergence criteria affect the result.

Task 4 - Assumption Register

The true historical movement of stockpiled material is not directly observed; the reconstruction is therefore non-unique and requires engineering assumptions. Maintain an explicit Assumption Register throughout the project.

- Bulk density and/or density by material type
- Swell, compaction, and settlement
- Angle of repose
- Dozer push direction and effective push distance
- Degree of mixing
- Accuracy of truck GPS coordinates
- Survey precision
- Within-load grade homogeneity
- Treatment of elevation decreases
- Material segregation
- Moisture effects, if relevant
- Missing-data treatment
- Numerical resolution and boundary conditions

For each significant assumption, state	Required content
1. Assumption	The value, rule, simplification, or model choice used.
2. Need	Why an assumption is required given the available information.
3. Justification	Engineering, operational, empirical, or literature-based rationale.
4. Consequence	How results could change if the assumption is incorrect.
5. Treatment	Whether it is tested through sensitivity/uncertainty analysis or otherwise controlled.

Important: "Assumed for simplicity" is not, by itself, sufficient engineering justification.

Task 5 - Development of the Stockpile Block Model

Use the reconstructed placement history to create a three-dimensional block model of the current stockpile. At minimum, each model block or equivalent spatial unit should contain appropriate estimates of:

- Block coordinates / spatial index
- Material volume
- Tonnage
- Copper grade
- Gold grade
- Contained copper
- Contained gold
- An appropriate measure, range, classification, or indicator of model uncertainty

Justify block dimensions, discretization/sub-blocking, partial-block treatment, density assignment, grade compositing, mixing rules, and treatment of uncertain or incompletely informed regions.

The final model should be usable, at least conceptually, for reclaim sequencing, blending, plant-feed forecasting, metal-production forecasting, and economic evaluation.

Task 6 - Model Validation and Reconciliation

A model is not considered reliable merely because the code runs. Develop a quantitative validation strategy.

Validation area	Minimum expectation
Geometry	Compare reconstructed and surveyed surfaces using elevation/volume differences and spatial error maps or equivalent metrics.
Mass balance	Compare cumulative FMS tonnage with reconstructed stockpile tonnage; quantify and explain discrepancies.
Metal balance	Compare Cu and Au entering the stockpile with the metal inventory represented by the final model.
Temporal behaviour	Where practical, test the algorithm on periods not used for parameter calibration or method development.
Robustness	Demonstrate that key results are not artifacts of a single arbitrary numerical setting.

Define quantitative performance measures appropriate to your method. Visual agreement alone is insufficient.

Task 7 - Uncertainty and Sensitivity Analysis

Do not present the final model as an exact representation of reality. Identify the principal sources of uncertainty and evaluate the effect of the most influential ones.

- Truck GPS error

- Survey error
- Tonnage measurement error
- Assay uncertainty
- Density uncertainty
- Unknown dozer trajectories
- Unknown degree of mixing
- Uncertain movement distance
- Uncertain angle of repose
- Missing data
- Algorithmic assumptions
- Spatial/numerical resolution

Your final presentation of results should distinguish, wherever practical, between measured information, inferred information, assumed information, and model-generated information.

Task 8 - Automated In-House Platform

The mine requires a repeatable process, not a one-time analysis. Develop a procedure or prototype in-house software platform that can update the analysis as new data becomes available.

You may use any appropriate programming environment, commercial software, open-source package, database, GIS/mine-planning tool, AI system, or combination of tools. A graphical user interface is optional. Engineering quality, reproducibility, and auditability are more important than visual sophistication.

1. Import new FMS records.

2. Import new survey surfaces.

3. Perform and report data-quality checks.

4. Identify new material placement and the relevant modelling period.

5. Apply the material-movement algorithm.

6. Update the stockpile block model.

7. Update tonnage and metal inventories.

8. Update uncertainty information.

9. Produce QA/QC and reconciliation outputs.

10. Generate appropriate visualizations and export files.

11. Retain sufficient records to audit assumptions, manual interventions, software versions, and previous model updates.

Reproducibility standard: A technically competent mine-planning employee who did not develop your project should be able to understand the workflow and reproduce the analysis using your submitted documentation and files.

6. Professional Practice Requirements

6.1 Use of Software, Code, and Artificial Intelligence

Your group may use AI tools. Students remain professionally responsible for the submitted engineering work. Acknowledge all substantive software, libraries, external algorithms, datasets, code sources, AI systems, and other external resources appropriately.

- AI system/tool used
- Purpose for which it was used
- Nature of the assistance obtained
- How the output was checked or validated
- Which technical decisions remained the responsibility of the team

Unverified AI-generated code, equations, citations, technical statements, or interpretations are not acceptable engineering practice.

6.2 Professional Accountability, Ethics, and Equity Scenario

> **Scenario:** Management requests a definitive copper and gold inventory and a reclaim sequence for next year. Your team knows that some regions of the stockpile have substantial uncertainty. A production target would be easier to justify if you presented only expected values and did not emphasize uncertainty. In addition, parts of the historical dataset are less complete for particular operating periods, crews, contractors, or data-collection systems, so model reliability is uneven across the stockpile.

Your team must determine how a professional engineering team should respond and what controls should be incorporated into the modelling workflow.

Professional accountability considerations

- Information and uncertainty that must be communicated to decision makers
- Limitations that must not be hidden or minimized
- Responsibility for validation and approval
- Required level of evidence before operational use
- Circumstances in which a definitive recommendation should be qualified or declined
- Documentation of assumptions, corrections, manual interventions, and model revisions
- Verification of AI-generated or externally sourced code before implementation

Equity and inclusive decision-making considerations

- Potential systematic bias from missing or unequal-quality data
- Risk of incorrectly attributing model discrepancies to particular crews, contractors, or groups
- Explainability of automated recommendations to people affected by them
- Inclusion of relevant perspectives from planning, geology, survey, operations, processing, and other users
- Usability by people with different levels of programming/modelling expertise
- Controls that promote fair, transparent, and responsible use of the platform

7. Teamwork Requirements

The project will normally be completed in teams of 3-4 students. Effective teamwork is an assessed component; a common technical submission does not by itself demonstrate individual achievement of the teamwork graduate attribute.

7.1 Team Charter

- Team members and initial roles
- Communication expectations and channels
- Decision-making process
- Procedure for resolving disagreements
- Meeting and contribution expectations
- Process for internal technical review and quality control
- Expectations for respectful and inclusive participation

7.2 Contribution Record

Maintain a concise record of major meetings, decisions, task assignments, completion of assigned responsibilities, and significant changes in responsibilities. This is not intended to be an hourly timesheet; it should show how the team coordinated and managed the work.

7.3 Individual Accountability

Although the principal technical work is a team product, every student is individually accountable for understanding the overall methodology. During the presentation or technical defence, any team member may be asked to explain the data treatment, assumptions, algorithm, validation, uncertainty analysis, software operation, or professional decisions.

8. Required Deliverables

Deliverable	Minimum content
D1. Team Charter and Work Plan	Concise document defining responsibilities, communication, decision making, conflict resolution, quality review, and inclusive participation.
D2. Data and Assumption Register	Dataset inventory, QA/QC findings, preprocessing decisions, excluded/corrected data, assumptions, justifications, consequences, and sensitivity treatment.
D3. Technical Methodology and Algorithm	Detailed description with equations, pseudocode/flowcharts, assumptions, numerical procedures, and engineering justification sufficient for independent understanding.
D4. Stockpile Block Model and Model Outputs	Model files and visualizations showing geometry, material distribution, Cu, Au, tonnage, contained metals, and uncertainty.

Deliverable	Minimum content
D5. Automated Analysis Platform	Source code/model files, dependencies, input format, operating instructions, workflow, example execution, and audit/reproducibility documentation.
D6. Final Engineering Report	Professional report to mine management integrating methodology, results, validation, uncertainty, limitations, ethics/equity, and recommendations.
D7. Team Presentation and Technical Defence	Professional presentation emphasizing the algorithm, assumptions, validation, uncertainty, software, limitations, and implementation recommendations. Every member participates.
D8. Individual Teamwork and Accountability Submission	Individual reflection plus confidential peer/self-assessment demonstrating contribution, coordination, leadership/followership, conflict resolution, workload equity, and inclusive team practice.
D9. Individual Professional Responsibility, Ethics, and Equity Assessment	Individual response to the project scenario identifying ethical issues, affected parties, uncertainty disclosure, risks, bias/equity concerns, professional responsibilities, required actions, and a justified recommendation.

9. Recommended Final Report Structure

1. Executive Summary
2. Introduction and Project Objectives
3. Description of Available Data
4. Data Analysis and QA/QC
5. Stockpile Growth Analysis
6. Material-Movement Methodology
7. Assumption Register
8. Block Model Construction
9. Validation and Reconciliation
10. Uncertainty and Sensitivity Analysis
11. Automated Platform
12. Results and Discussion
13. Professional Accountability, Ethics, and Equity
14. Limitations
15. Recommendations for Mine Implementation

16. Conclusions
17. References
18. Appendices

10. Assessment Structure

Component	Weight
Data analysis, QA/QC, and assumptions	10%
Material-movement algorithm and engineering justification	25%
Block model, validation, and uncertainty	15%
Automated in-house platform	20%
Engineering report and presentation	10%
Individual and Team Work - Group Teamwork	10%
Ethics and Equity - Accountability and EDI	10%
Total	100%

The instructor may adjust the administrative distribution of points among individual submission items while preserving the overall technical and graduate-attribute expectations stated above.

11. Standard of Engineering Quality

The strongest project is not necessarily the one with the most complex algorithm. High-quality work demonstrates sound engineering reasoning, defensible assumptions, appropriate use of data, mass and metal conservation, quantitative validation, transparent treatment of uncertainty, reproducibility, effective teamwork, professional accountability, and clear recognition of methodological limitations.

Central expectation: Develop a defensible engineering methodology for reconstructing and managing a strategically important mine stockpile when the true historical movement of the material cannot be directly observed.

12. Submission Checklist

☐ Team Charter and Work Plan submitted.
☐ Data/Assumption Register included.
☐ Algorithm documented with sufficient technical detail.
☐ Mass and metal conservation checked.
☐ Stockpile block model and export files included.

- ☐ Validation metrics and reconciliation results included.
- ☐ Uncertainty/sensitivity analysis completed.
- ☐ Automated platform and operating documentation included.
- ☐ AI/software use fully disclosed and verified.
- ☐ Professional accountability and EDI requirements addressed.
- ☐ Individual teamwork reflection and peer/self-assessment submitted.
- ☐ Individual ethics/equity assessment submitted.
- ☐ All files are named clearly and can be opened by the instructor.